Course Project Report
Group 10

Submission date:

We declare that:
(i) all members of the group have read and checked that all parts of the project (including proposal, code programs, and reports), irrespective of whether they are contributed by individual member or all members as a group, here submitted is original except for source material explicitly acknowledged;
(ii) the project, in parts or in whole, has not been submitted for more than one purpose without declaration; (iii) we are aware of the University's policy and regulations on honesty in academic work and understand the possible consequence when breaching such policy and regulations;
(iv) we confirm that we have declared in the report about the usage of AI and other generative models, including but not limited to ChatGPT, LLaMA, Gemini, Mistral, and Stable Diffusion, and complied with the instructions provided by HKMU; and
(v) we are aware that all members of the group should be held responsible and liable to disciplinary actions, irrespective of whether he/she has signed the declaration and whether he/she has contributed, directly or indirectly, to the problematic contents.

We confirm that we have read through and understood the project requirements. We understand that failure to comply with the project requirements will result in score deduction.

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Inventory system:

https://github.com/yuf44643-ai/inventory_system/tree/main

Video Link:

(task1)https://youtu.be/CZ5dlygkbJk

(task2)https://youtu.be/FXCunv-_brU

1.Introduction:

Our project implements a graphical inventory management system. It is using Python and Tkinter, extended with a secure login module and permission control. The system supports three user roles: Admin, Manager, and Staff.

1.1 Core Function:

(1)User Module: A login interface verifies username and password, identifying user role, granting corresponding permissions.

(2)Inventory Operations: Add, Remove, Update, View product information.

(3)Role Access Control:

*Admin: Full access to all functions.

*Manager: allowed to add and remove products.

*Staff: Only allowed to view the inventory.

(4)Data Validation and Error Handling: Prevents invalid inputs such as empty fields, duplicate product IDs, and negative price or quantities.

(5)Graphical User Interface(GUI): Clean, user-friendly layout with responsive button states based on user permission.

1.2 Techniques Used:

* Python as the main programming language.

* Tkinter and ttk for building a modern, styled GUI.

* Object-Oriented Programming(OOP) for modular and maintainable code structure.

* Role-based permission logic using class and method.

* Separation of concern: login, user, product, inventory, and GUI are split into independent modules.

2.Code Section:

2.1 Class and Object:

(1)Product: Represents a product with attributes (ID, name, price, quantity)

(2)Inventory: Manager a collection of products (add, remove, update)

(3)User: (AdminUser, ManagerUser, StaffUser)--->Different users have different rights.

(4)LoginGUI & InventoryGUI: Handle graphical interface logic.

2.2 Encapsulation:

*Private attributes (e.g. _username, _role) are used to protect internal state.

*getter and setter methods control access to sensitive data, preventing direct modification.

2.3 Inheritance

*AdminUser, ManagerUser and StaffUser inherit from the base User Class.

*Common properties and method are defined in the parent class.

* This avoids code duplication and simplifies future role extensions.

2.4 Polymorphism

*Different user classes override method such as generate_full_report() to provide role output.

*The permission check method can_perform_action() behaves differently based on the user role object passed into GUI.

2.5 Modularity and Abstraction:

*Functions are grouped into separate modules (login.py, user.py, product.py, inventory.py, gui.py, main.py).

*Complex logic is hidden behind simple method calls, making the system

easy to understand and maintain

3.External Resources:

No external third-party libraries or external codebases are used beyond Python's built-in modules:

* tkinter and tkinter.ttk for GUI development.

*typing for optional type hints to improve code easily to read.

*messagebox from Tkinter for user notifications.

All code is independently designed and implemented by the author. No code from other programmers, online tutorials, or open-source projects is directly copied.

4. Potential Weaknesses and Future Improvements:

Weaknesses:

(1):User are stored in plaintext within the code, which is insecure for real application.

(2):No persistent data storage. All product data resets when the program closes.

(3):The interface supports only styling and is not fully responsive across different screen sizes.

(4):Limited error handling for edge cases such as extremely large input values.

Future Improvement:

(1):Store user accounts and product data in a database such as SQLite or JSON files for persistence.

(2):Encrypt password using hashing libraries like bcrypt or hashlib.

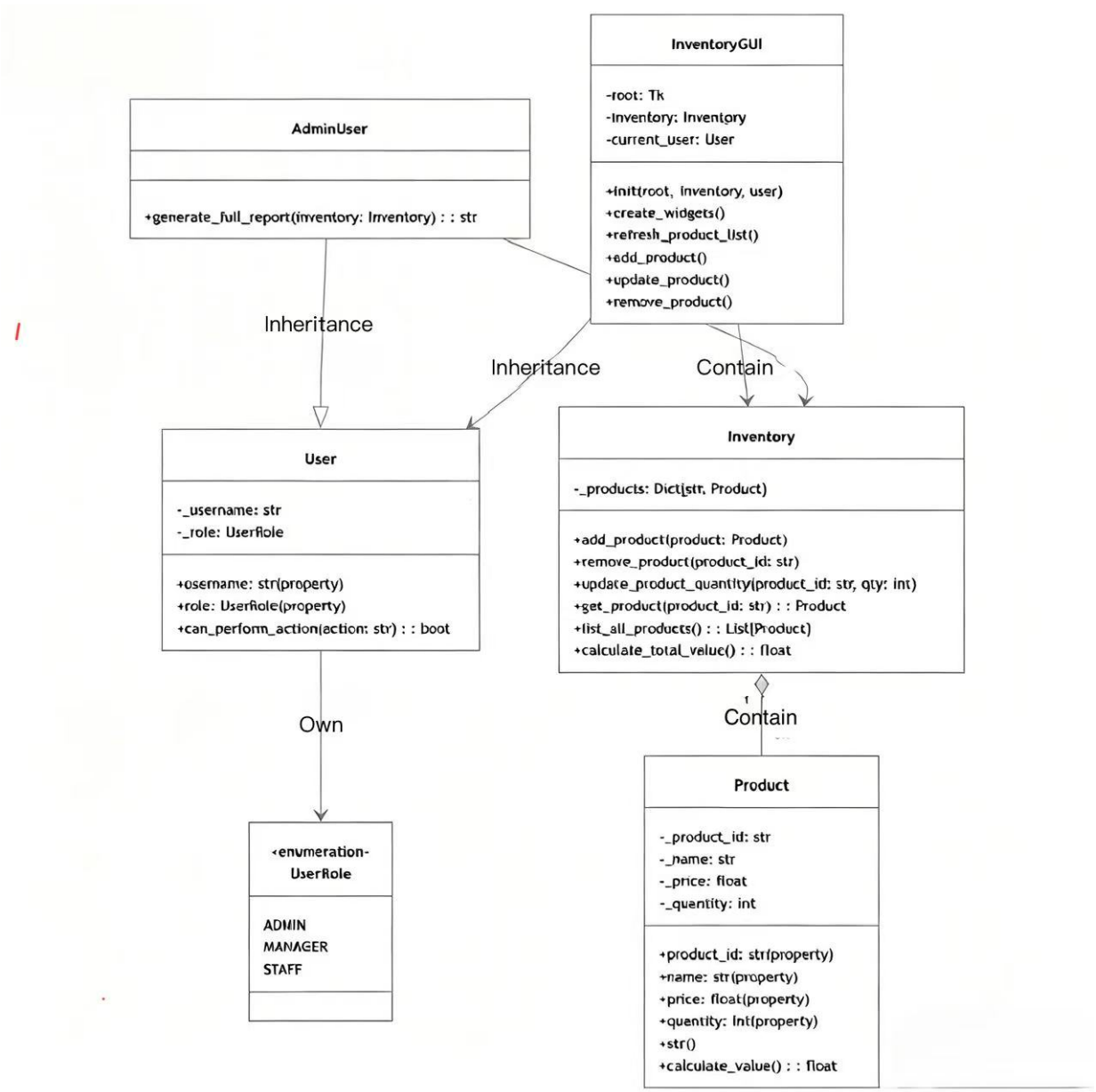
(3):Add data export functions (CSV,Excel) for inventory for inventory

reports.

(4):Implement a more modern GUI

5. Reference:

1.https://docs.python.org/3/library/functions.html

2.https://www.w3schools.com/python/

Appendix: